

Drivers application. You should see it enabled and working—or, at the very least, it should show up, and you can click *Activate* to get it going.

In Fedora, the procedure is simpler. Simply install the RPMFusion repositories, and type in the code below:

```
sudo yum makecache all
sudo yum -y install akmod-wl
```

And in case you use openSUSE, the drivers are again available in the Packman repositories. Install the repositories, and then install the *broadcom-wl* and one of the *broadcom-wl-kmp-Skernelflavor* packages.

If all else fails and you need to build from source, the instructions on how to obtain, build and install the drivers are at: http://www.broadcom.com/docs/linux_sta/README.txt

Power management

Power management in Linux is nothing short of the devil's playground. In Linux, the battery drains twice as fast as on Windows 7 (from 6 hours on Windows 7 to 3 hours in Ubuntu). To maximise your battery life, you should probably perform the following steps.

Get to the *Power Management* applet (*System->Preferences->Power Management*) and then go to the *On Battery Power* tab. The optimal settings are somewhat as follows:

- *Put Computer To Sleep When Inactive For: 10 Minutes.*
- *When Laptop Lid Is Closed: Suspend*
- *When Battery Level Is Critically Low: Hibernate*
- *Spin Down Hard Disks When Possible: Checked*
- *Put Display To Sleep When Inactive for: 5 Minutes*
- *Reduce Backlight, and Dim Display When Idle: Both Checked*

Then click *Make Default*, and you're good to go.

A command-line tool that you'd want to install is *PowerTOP*. It's a tool like *top*, which shows processes and processor usage, but with a twist—this actually measures processor wake-ups from idle, and power consumption per process.

Bluetooth

The default Bluetooth applets that come with both GNOME and KDE are too crippled. You'd generally need to install a package called *Blueman*, which is much more powerful, has a better interface, a *Device Browser*, and can make your computer discoverable or turn it off. Besides, re-connecting the already-paired Jabra BT2015 was impossible without *Blueman*.

Miscellaneous stuff

You'd have to disable desktop compositing when on battery. However, I could not find a way to do that automatically. You'd also want to conserve on any unnecessary daemons running while on battery, but that'll require diving into quite a lot of configuration files.

And last but not the least, almost all laptops use Windows-specific dial-up modems. Pretty stupid, I know, but they just don't work under Linux—period. Dell does provide the HSFModem driver from Linuxant, but it's for older kernels only.

If you use a USB modem like Tata Photon+ or Reliance NetConnect to get on the Internet, the way to do it is as follows:

As soon as you insert the modem, it should be recognised as a USB serial device, and you should have a */dev/ttyUSB0* node added automatically. Now, in a terminal window, type *pppconfig* to start the PPP Configuration utility. For both, the *Phone Number* should be *#777*, *IP Configuration* should be *Dynamic*, and *Authentication Mode* should be *PAP*. The only difference is in *Username* and *Password*: For Reliance NetConnect, this is the phone number of the device, while for Tata Photon+ it is 'internet'.

To start and stop these connections, get to a prompt and type in the following:

```
sudo pon <your connection name> # to start the interface
sudo poff <your connection name> # to stop the interface
```

If you are using a cellphone (GPRS or EDGE connection), via Bluetooth, you are in for some luck. Start the *Blueman Device Manager*, right-click on the entry for your phone, and click *Setup*. Select *Dial-Up Networking* and proceed. Once that's done, it's time to set up the actual connection.

Click on the *NetworkManager* icon, and click *New Mobile Broadband (GSM) Connection*. A wizard starts up. Click *Forward*. Now select your country (*India*), and click *Forward*. Select your *Provider* (a pleasant surprise is that all major service providers, including Tata DoCoMo, are supported). Anyway, I selected *Airtel*, hit *Forward* twice, and clicked *Apply*. In moments, I was connected. To connect or disconnect, just click on the *NetworkManager* icon and choose the appropriate option.

Well, that's all for this article. Hope that with the help of this you are able to make some headway in your laptop adventures with Linux. Till the next time... 

Resources

- How To Build Kernel Packages For Almost All Distros: <http://www.howtoforge.com/howtos/linux/kernel>

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